

Goal

The purpose of the technical interview is to assess the candidate along these dimensions

- Ability to design solutions that are easy to understand and practical to implement
- Ability to apply AI/ML to solving real world problems
- Ability to prototype solutions to a point that they can be used to validate product and engineering requirements
- Collaboration skills & style while designing and developing an application with other engineers

Prompt

Collaboratively design and prototype an application that leverages Large Language and Machine Learning Models to optimize a tractor manufacturer's supply chain by predicting demand fluctuations, supplier delays, and component failures, and recommending cost-effective inventory strategies.

Optimize - ensure you always have the right number and type of supplies to create enough tractors to meet demand without any excess

Supply Chain - The companies that supply you with parts to build tractors

Demand Fluctuations - The demand of customers for new tractors

Supplier Delays - delay in manufacturing and delivery of a part for a tractor

Component Failures - Broken or critical defect of a part that prevents it being used for manufacture, or breaking of the component while the tractor is being used

Stage 1

90 Minute System Design

- Candidate and Kontakt.io engineering interviewer collaboratively review the solution requirements and iterate on a design for implementing the solution
- Output
 - System Design Diagram
 - API Definitions
 - UI Mockup

- Identification of models, their data sources, inputs and outputs